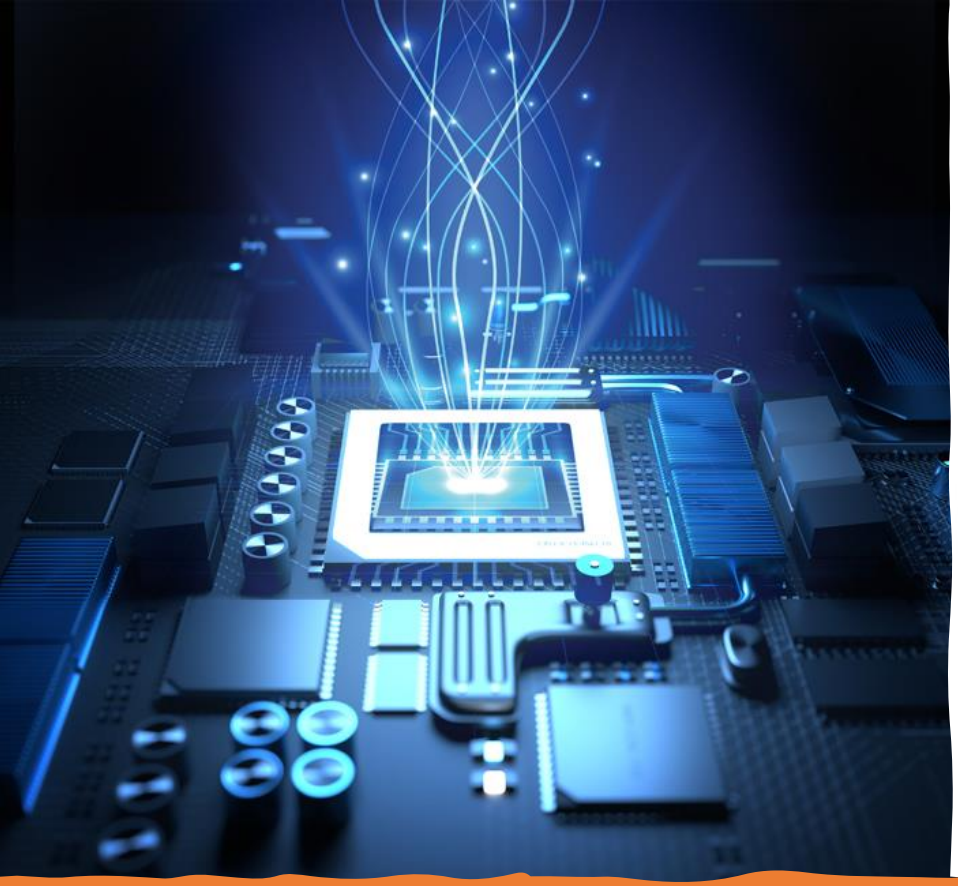
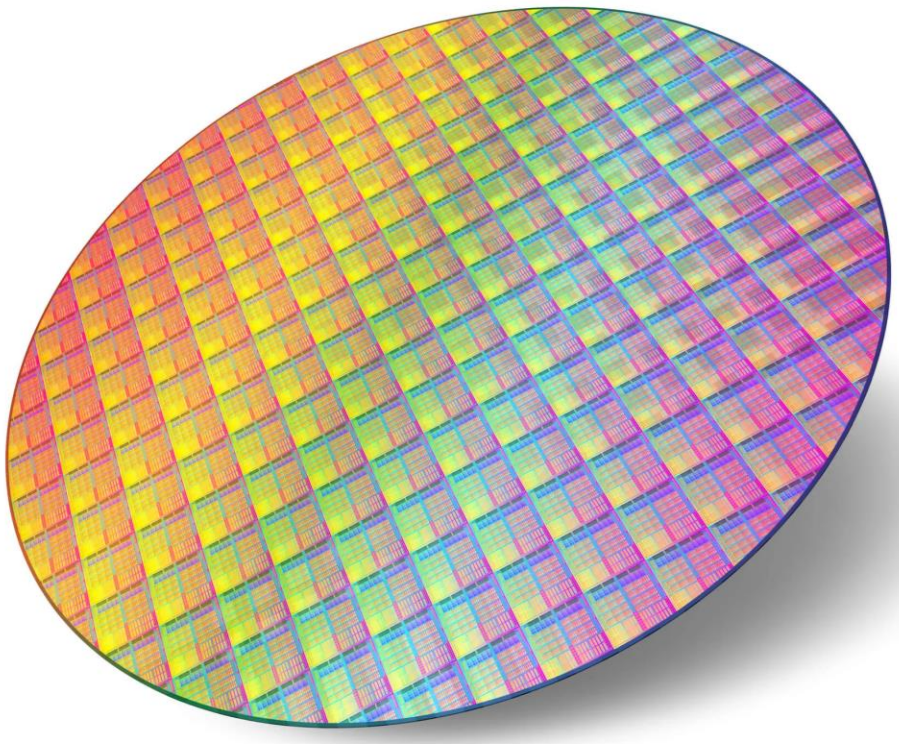




EE 4121
VLSI Design and Technology
Credit: 3.00

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Course Contents

Introduction to microelectronics and MOS technology, basic electrical properties and circuit design processes of MOS and Bi-CMOS circuits, Layout design rules, and sub-system design processes.

Computational elements: design of CPU and ALU sub-system, data path, adder, multipliers, memory, registers, and aspects of system timing, practical aspects of design tools and testability

Device modeling: MOS, diode, bipolar and passive component; CMOS Design: Behavioral description, structural description, physical description, and design verification.

Fabrication and processing technology: Crystal growth, wafer preparation, photolithography, oxidation, diffusion, ion implantation, epitaxial metallization, etching, NMOS and CMOS fabrication technology.

Testing and packaging: Overview of silicon semiconductor technology, power dissipation, packaging, and silicon on insulator. Introduction to GaAs technology: Ultra-fast VLSI circuits and systems.

Course Objectives

- ✓ Design MOS circuits using the principles of circuit design
- ✓ Create models of MOS, diode, bipolar, and passive components
- ✓ Run HDLs and simulate using FPGAs

Course Outcome (CO)

CO1	**Analyze mask layout, DRC, and LVS of VLSI circuits**
CO2	Design different logic gates and basic VLSI building blocks for complete system design
CO3	Apply HDL to design, synthesize, and implement a digital system

Reference Books

1. VLSI Technology - S. M. SZE
2. Silicon VLSI Technology: James D. Plummer, Michael Deal, Peter B. Griffin
3. Semiconductor Device Physics and Technology - S. M. SZE
4. Basic VLSI Design: Pucknell
5. Principle of CMOS VLSI Design – Neil H. E. Weste, Kamran Eshraghian
6. Design of VLSI Systems: A Practical Introduction: Linda E. M. Brackenbury

What is VLSI?

- **VLSI (Very Large Scale Integration)** is the process of integrating millions or even billions of transistors onto a single integrated circuit (IC) chip

Key Features:

- High transistor density
- Compact chip design
- High performance
- Low power consumption
- Cost-effective manufacturing

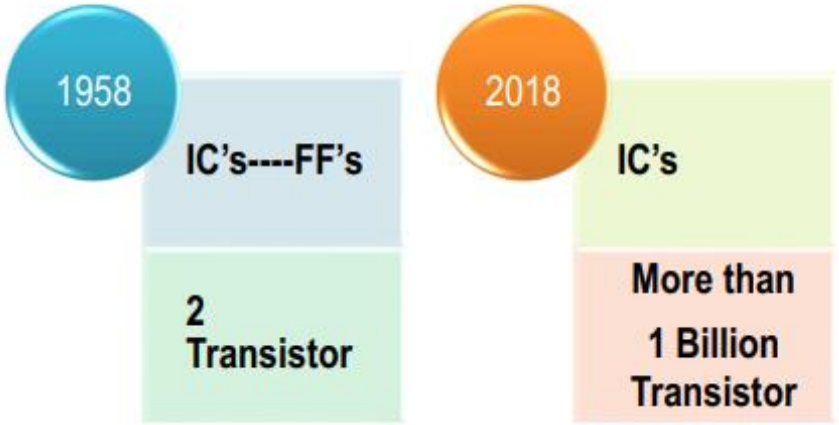


Evolution of VLSI

- SSI (Small Scale Integration): 10–100 transistors
- MSI (Medium Scale Integration): 100–1,000 transistors
- LSI (Large Scale Integration): 1,000–10,000 transistors
- VLSI (Very Large Scale Integration): Millions to billions of transistors
- ULSI (Ultra Large Scale Integration): Advanced modern chips

Scale of Integration

► **Very-Large-Scale Integration (VLSI)** refers to the process of designing and fabricating integrated circuits (ICs) by integrating hundreds of thousands to billions of transistors and other electronic components into a single semiconductor chip.



THE PROGRESS ON PROCESSORS

Intel processor and introduction year	Frequency	Number of transistors	Smallest dimension
4004, 1971	108 kHz	2,300	10 µ
8080, 1974	2 MHz	4,500	6 µ
286, 1982	6 MHz	134,000	1.5 µ
486, 1989	25 MHz	1,200,000	1 µ
Pentium Pro, 1995	200 MHz	5.5 million	0.6 µ
Pentium III, 1999	500 MHz	9.5 million	0.18 µ
Pentium M,	1.7 GHz	55 million	90 nm
Core 2 Duo, 2006	2.93 GHz	291 million	65 nm
Penryn, 2007	>3 GHz	820 million	45 nm
Next generation, late 2008	...	1.9 billion	32 nm

Dual Core+ Core I7 (2015): 1.9 billion and 22 nm

➢ **This scale of growth has resulted from a continuous scaling of transistors and other improvements in the Silicon manufacturing process.**

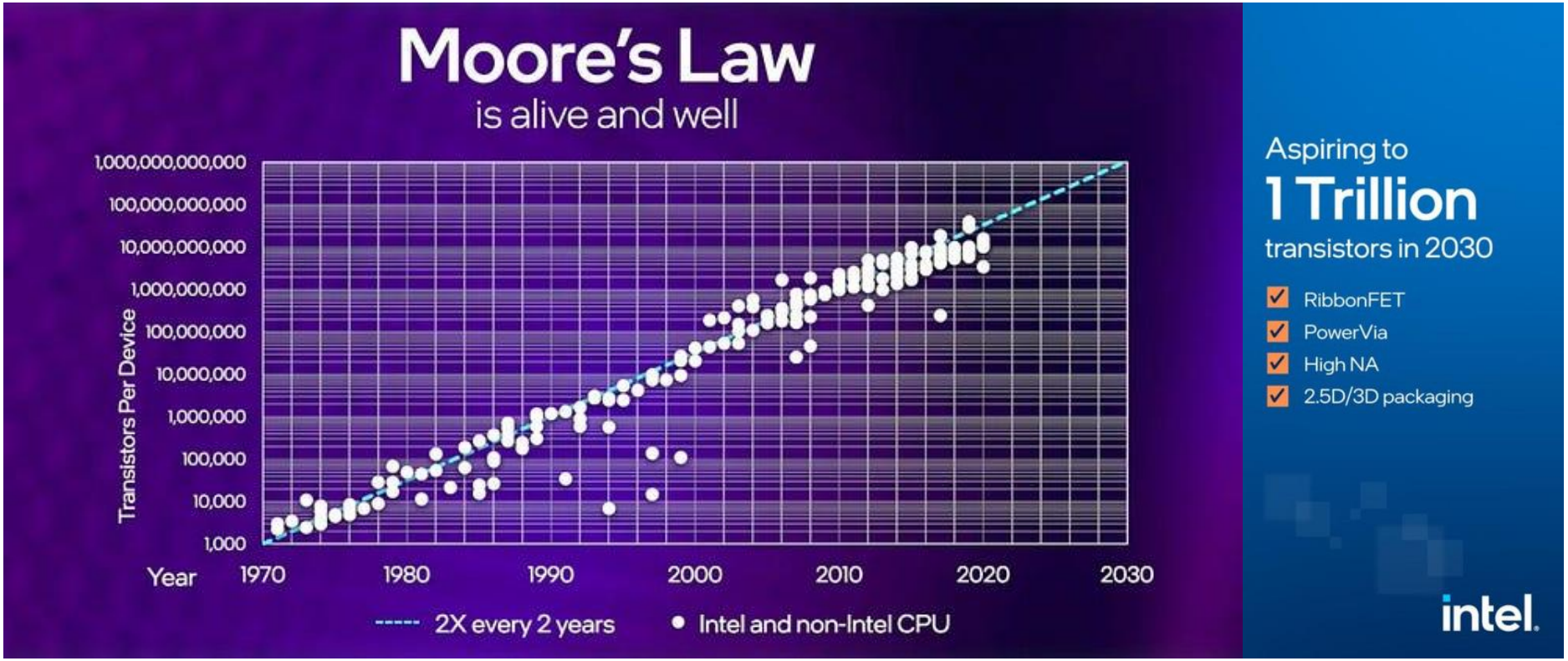
How many transistors are in the Intel Core i9 or Core Ultra 9 Processor?

- **Modern Core i9:** about **20–25 billion transistors**, depending on the model

- **Core Ultra 9 - Not Public**

Moore’s Law

- **1965: Gordon Moore** (Intel cofounder) plotted transistor on each chip
- **Moore's Law states that the number of transistors on a microchip approximately doubles every two years, leading to exponential growth in computing performance while reducing the cost per transistor**



Importance/ Advantages of VLSI Technology

- Enables miniaturization of electronic devices
- Increases processing speed
- Reduces power consumption
- Improves system reliability
- Lowers manufacturing costs
- Supports advanced computing applications

Applications of VLSI

- Smartphones and tablets
- Computers and laptops
- Artificial Intelligence (AI)
- Internet of Things (IoT)
- Automotive electronics
- Medical devices
- Communication systems
- Consumer electronics

Challenges in VLSI

- Heat dissipation
- Design complexity
- Manufacturing cost
- Testing and verification
- Signal integrity issues
- Security concerns

Chip Manufacturing Companies

Company	Country	Specialty
TSMC	Taiwan	World's largest foundry
Samsung Electronics	South Korea	Memory & logic chips
Intel	USA	CPUs & AI chips
GlobalFoundries	USA	Specialty semiconductor manufacturing
SMIC	China	Semiconductor foundry

Chip Design Companies

Company	Main Products
NVIDIA	GPUs & AI chips
AMD	CPUs & GPUs
Qualcomm	Mobile processors
Apple	A-series & M-series chips
Broadcom	Networking chips

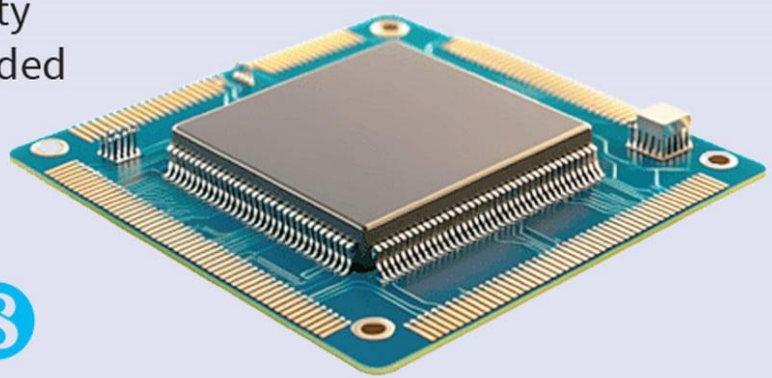
Bangladesh in the Global Semiconductor Market

- Bangladesh targets chip design, testing, packaging; avoids costly fabrication initially
- Government extends semiconductor tax incentives and duty cuts until 2031
- Goal: grow semiconductor exports from $8 million to $1 billion
- Bangladesh produces thousands of engineering graduates supporting semiconductor talent growth
- Local firms already provide international chip design and engineering services
- Skills training, infrastructure, and research investment remain major challenges

https://www.tbsnews.net/bangladesh/why-bangladesh-charts-design-first-path-semiconductors-ditching-wafer-fabrication-1469936

WHY BANGLADESH IS CHOOSING A NICHE SEMICONDUCTOR STRATEGY

GOVERNMENT PUSH IN FY27 BUDGET | VAT & customs duty exemptions extended until June 2031



Exemptions cover

- Electronic Design Automation tools
- Chip design software
- Packaging machinery
- Testing equipment



It wants to avoid capital-heavy wafer fabrication

TALENT ADVANTAGE

- **13,000** EEE graduates/year
- **26,000** CSE graduates/year

POLICY & ECOSYSTEM GAPS

Experts call for



Semiconductor testing facilities

- Packaging plants
- Stronger IP protection
- University-industry collaboration
- Easier access to incentives, Hi-Tech Park land

The End